

OMER AKBEN

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PROFESSIONAL SUMMARY

AI Full-Stack Engineer and EdTech Founder with 6+ years of SDET/QA automation experience. Creator of [TUEL AI](https://tuel.ai), a B2B educational technology platform deployed at Elon University with production metrics demonstrating real user impact. Specializes in building production-grade Retrieval-Augmented Generation (RAG) and agentic workflows using LangGraph, LangChain, and modern LLM APIs (OpenAI, Claude, Vertex AI). Combines strong Python (FastAPI) and TypeScript/React (Next.js, Vercel AI SDK) skills with deep testing and CI/CD expertise to ship reliable, capability-building AI applications. Focused on human-AI interaction that develops genuine user skill rather than engagement metrics. Known for system-level thinking: observability, automated evaluations (RAGAS, DeepEval), and shift-left quality baked into full-stack delivery.

CORE SKILLS

Languages: Python, TypeScript, JavaScript, C#, Java, SQL, Bash

AI & LLM: LangChain, LangGraph, OpenAI API, Anthropic Claude, Vertex AI, Azure AI / Microsoft Foundry, Amazon Bedrock, RAG, RAGAS, LangSmith, Arize Phoenix, vector databases (pgvector, FAISS, Pinecone, Weaviate), multi-agent systems (CrewAI, Mastra), prompt engineering, golden datasets

Backend & APIs: FastAPI, Django, Node.js/Express, REST, GraphQL, Pydantic, OpenAPI/Swagger, OAuth2/JWT, ORMs, API versioning

Frontend: React, Next.js (App Router), Vercel AI SDK, streaming UIs (SSE), Tailwind CSS, SSR/ISR, accessibility (a11y), performance optimization (Lighthouse)

Data & Infra: PostgreSQL, SQL Server, MongoDB, Redis (semantic caching, rate limiting), Elasticsearch, schema design, migrations, query tuning

Cloud, DevOps & LLMops: AWS (Lambda, ECS, S3, API Gateway), Azure (Entra ID, DevOps), Docker, Kubernetes, GitHub Actions, Jenkins, Terraform, observability, cost/performance optimization

QA & Test Engineering: Playwright (E2E/API), Selenium WebDriver, Cypress, REST Assured, Postman/Newman, Cucumber/SpecFlow (BDD), JUnit/TestNG/xUnit/MSTest, JMeter, Appium, test strategy, TDD, BDD, shift-left practices

PROFESSIONAL EXPERIENCE

Founder & AI Full-Stack Engineer — TUEL AI (tuel.ai) | Remote (Raleigh, NC) | Jan 2024 – Present

Clients: [Elon University](#), [Minor Use Foundation](#), [North Glass LLC](#)

- **Selected for Elon University Business Faculty Pilot Program** after 6 months production use. Built AI education platform with 16+ interactive learning tools (flashcards, quizzes, concept maps, Hume AI voice chat) and 7-layer prompt system with source citations. **Production metrics:** 100+ sessions, 50+ users, 20M+ tokens, 2% error rate, 95% satisfaction (62/65 positive), 60% support response time reduction.
- Built FERPA-compliant multi-tenant architecture with row-level isolation, Microsoft Entra ID RBAC, faculty dashboard for AI behavior control, and 35+ audit action types. Implemented production RAG pipeline (pgvector + keyword hybrid search, Upstash Redis semantic caching, RAGAS validation) achieving 25% retrieval precision improvement with P95 latency under 2 seconds.
- Orchestrated multi-agent workflows in LangGraph and AI SDK for autonomous tool calling and student services flows with content guardrails and conversation memory. Automated regulatory document workflows for Minor Use Foundation using LLM-based PDF extraction, reducing manual data entry by 60% with full audit trails.
- Integrated Playwright E2E tests into GitHub Actions and LangSmith tracing, achieving 95%+ stability on critical chat/auth flows and cutting query costs by 15%.

Test Automation Architect (SDET) — Oteemo | Remote | Feb 2025 – Sep 2025

- Led quality engineering strategy for Fortune 500 clients, transforming greenfield Azure DevOps into a unified automation ecosystem. Designed containerized (Docker) Playwright/Selenium infrastructure with environment parity, cutting project setup from days to hours.

- Built deterministic test patterns (intelligent retries, self-healing locators) and parallelized Jenkins/Azure pipelines, achieving 98% pass rates and 40% faster PR feedback. Mentored 5+ engineers on TDD/BDD and shift-left practices.

QA Engineer — Engineering Consulting Services (ECS) | Raleigh, NC (Hybrid) | Apr 2023 – Jan 2024

- Refactored core test utilities using Builder/Strategy patterns, increasing team velocity by 20% and reducing code duplication by 35%. Modernized legacy Selenium suites to Page Factory with resilient locators, boosting UI stability from 65% to 95%. Automated API validation with Postman/Newman in Azure DevOps CI.

SDET / QA Test Automation Engineer — Xsolis | Remote | Oct 2022 – Apr 2023

- Automated healthcare workflows (HL7 message processing, PHI ETL, clinical documentation) using Selenium/SpecFlow, increasing coverage from 40% to 85% on compliance workflows (PCI, SOX). Implemented SQL-based validation across 50+ tables with automated reconciliation reports. Collaborated with EPAM on shared xUnit/Postman suites.

SDET / QA Automation Engineer — Fannie Mae | Remote | Mar 2021 – Sep 2022

- Built Java-based automation framework (Selenium WebDriver, TestNG, REST Assured) for mortgage processing modules, achieving 20% automated coverage and 10% faster releases. Architected Selenium Grid CI infrastructure with 50+ concurrent browser sessions, reducing regression time by 15%. Implemented REST Assured API contract testing and AWS EC2/ECS deployment strategies.

QA Test Engineer — Delaware State Chamber of Commerce | Wilmington, DE (Hybrid) | Sep 2019 – Mar 2021

- Developed Java-based Selenium/JUnit framework integrated with Jenkins CI/CD, increasing automated coverage by 35%, reducing API defects by 30%, and cutting execution time by 10%.

PROJECTS

AI Studio Platform Internal Project (Elon University) — Next.js/TypeScript • FastAPI • OpenAI/Claude/Gemini • LangChain/LangGraph • RAG (pgvector/FAISS) • Microsoft Entra ID • Playwright

- **Elon AI** ([elon-ai.app](#)) — Live deployment of TUEL AI. Next.js 15, FastAPI, Vercel AI SDK, multi-model (OpenAI/Claude/Grok/Gemini), LangChain/LangGraph, RAG (Upstash Vector/pgvector), Entra ID SSO, Hume AI voice, Drizzle ORM, Neon Postgres, QStash, 18 generative UI tools. Multi-tenant platform with agentic workflows, RBAC, RAGAS evaluation, and Playwright E2E/API tests tracking accuracy, precision/recall, and latency.
- **Elon AI Toolbox** ([Elon AI Toolbox](#)) — Elon University Generative AI Toolbox
Discover our collection of powerful generative AI tools and custom chatbots designed for Elon University staff, faculty, and students to enhance productivity, support research, and elevate teaching and learning experiences across campus.

Portfolio ([omerakben.com](#)) — Next.js/TypeScript • Vercel AI SDK • Mastra • Upstash Redis • Playwright

- AI-native portfolio with Mastra agents, Vercel AI SDK, Upstash Redis. Quality gates (Vitest, Playwright, Axe-Core), Lighthouse CI + Sentry + PostHog monitoring, sustaining 95+ scores for Performance, Accessibility, and SEO.

tuel ([npm packages](#)) — React • TypeScript • Storybook • Email Automation • Analytics

- Open-source, tree-shakable React animation component library with type-safe APIs, Storybook docs, CI-based testing, and automated npm publishing.

North Glass ([North Glass LLC](#)) — React/Next.js • TypeScript • SEO • CI/CD

- Developed a custom, SEO-optimized Next.js application for customer preferences. Optimized for Core Web Vitals

EDUCATION & CERTIFICATIONS

Nashville Software School — Full-Stack Web Developer Bootcamp | 2024–2025

Modern web stack (React/Next.js, Python/FastAPI/Django, TypeScript, TDD, CI/CD, Agentic AI Tools for Developers, Cloud Deployment Fundamentals).

TechCenture Academy — Software Development Engineer in Test (SDET) | 2018–2019

Test automation training covering Java/C#, Selenium WebDriver, Cucumber/SpecFlow (BDD), TDD/BDD, and CI/CD integration with Jenkins and Azure DevOps.

Istanbul Okan University — Master of Science, Healthcare Management | 2014–2016